

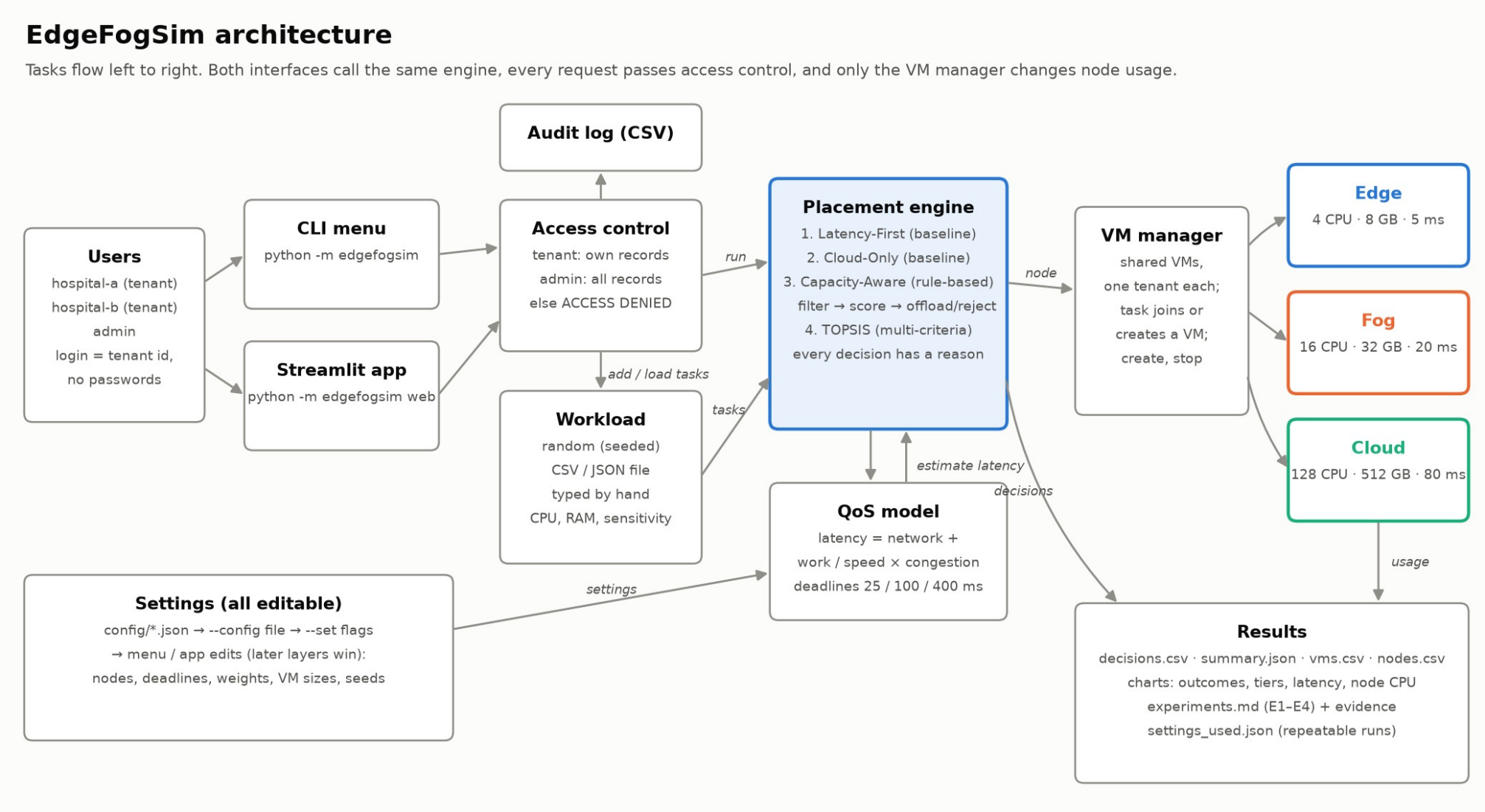
EdgeFogSim

Where should a task run: Edge, Fog or Cloud? A rule-based placement simulator with VMs, tenant security and QoS results.

ARM 305 – Cloud, Dev, Edge and Fog Computing · Mini Project

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Every task passes access control, a policy and a shared VM



Capacity-Aware: filter, score, then offload or reject

1. **Prefer tiers by urgency**

   high → Edge, medium → Fog, low → Cloud

2. **Filter out nodes**

   no room for the task's VM, CPU load over 85%, or the deadline would be missed

3. **Score the rest**

   0.5 × tier preference + 0.3 × time to spare + 0.2 × free CPU

4. **Place or reject**

   best score wins (offload if not first tier); if none fits, reject with the reason

Latency model

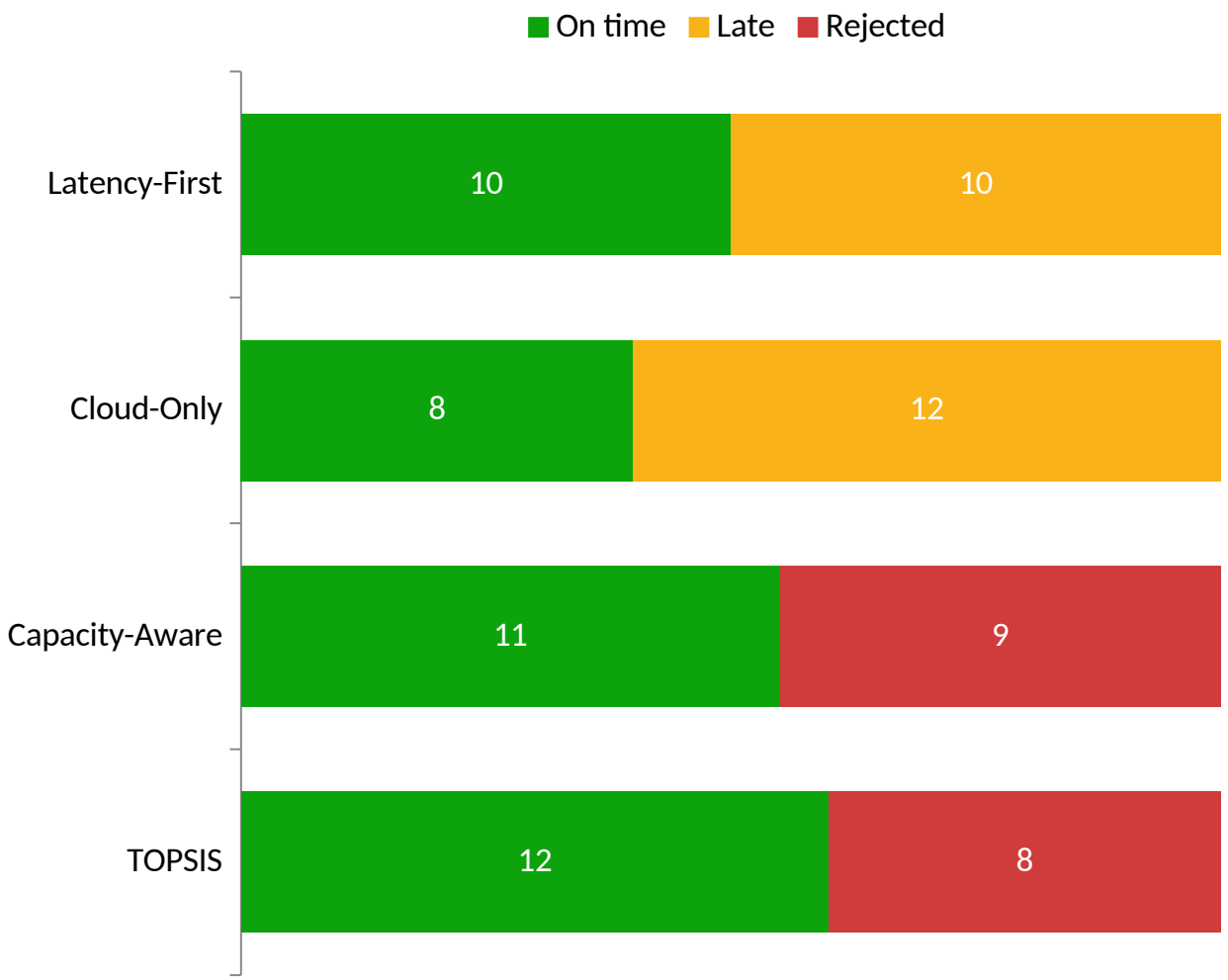
network + (work ÷ speed) × congestion
congestion = 1 ÷ (1 − CPU load)

Deadlines: high 25 ms · medium 100 ms · low 400 ms

Task	Sensitivity	Result
T01	high	edge-1, 18 ms
T02	high	fog-1 (offload), 25 ms
T03	low	cloud-1, 101 ms
T04	high	rejected

Sample tasks T01–T04 on empty nodes.

Smart policies run 0 late; baselines run 10 and 12 of 20 late



0

late with Capacity-Aware

10

late with Latency-First

20 generated tasks, seed 42 (11 high, 5 medium, 4 low sensitivity).

Overload is offloaded or rejected; other tenants are refused

E2 · Overload

Workload: 10 generated tasks, seed 7 (10 high sensitivity). Before the burst: edge-1 already 90% busy

4	6	0
offloaded to Fog	rejected, with a reason	late

Latency-First, same burst: 1 on time, 9 late.

No node is ever allocated beyond its capacity.

E3 · Unauthorised access

Logged in as hospital-b, trying another tenant's records:

DENIED	read task T01
DENIED	read VM vm-0001
DENIED	stop VM vm-0001

Each refusal is one line in the audit log; the VM keeps running; the admin can read both records (4/4 checks).

More Edge CPU helps; a slower Cloud pushes medium tasks out

edge-1 cpu_cores 4 → 8

Metric	Before	After
on time	11	12
rejected	9	8
avg latency ms	70.9	65.3
tasks on edge	1	2
tasks on cloud	6	6

✓ edge-1 takes at least as many tasks (1 → 2)
✓ At least as many tasks on time (11 → 12)
✓ No more tasks rejected (9 → 8)

cloud-1 base_latency_ms 80 → 150

Metric	Before	After
on time	11	9
rejected	9	11
avg latency ms	70.9	96.6
tasks on edge	1	1
tasks on cloud	6	4

✓ Tasks on cloud-1 are slower on average (104 → 179 ms)
✓ Low-sensitivity tasks all still finish on time (4 / 4)
✓ No task runs late (late 0 → 0)

What we learned

- All 4 experiments matched their expected outcome (19/19 checks)
- Checking the deadline before placing beats placing on the nearest node
- Shared tenant VMs and one access rule keep tenants apart
- Every number is a setting; runs are repeatable from settings_used.json

Limitations

- No task duration or arrival time; tasks run until their VM stops
- Latency is a formula, not a measurement
- Greedy, one task at a time; GA/PSO or Q-learning could do better
- No VM migration; login is a tenant id by design

Try it: python -m edgefogsim · python -m edgefogsim web · python run_experiments.py